Do Protests Really Spread?

Self-Excitation in Indonesian Protests

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Motivation

Research Question

Central question: Do protests really spread?

Scholars have long studied protest waves and diffusion, but a key puzzle remains:

- Does observed clustering reflect *true contagion*—protests causing protests?
- Or *common shocks*—independent responses to shared conditions?

Two sub-questions:

1. Do protests elsewhere predict more protests here?
2. If so, does geography matter—is spread local or national?

Prior Work on Protest Diffusion

Rich qualitative literature:

- Protest waves and cycles (Tarrow 1998; Beissinger 2002)
- Tactical diffusion (McAdam 1995; Soule 1997)
- Mechanisms: media, networks, demonstration effects

Limited quantitative evidence:

- Early event-history models (Spilerman 1970; Myers 2000)
- Cannot separate local vs. national spread
- Cannot quantify diffusion rate

The Identification Problem

Challenge: Protests cluster in space and time, but why?

Standard approaches struggle to distinguish:

1. **Common shocks** — National events (elections, price spikes) trigger protests everywhere simultaneously
2. **Correlated fundamentals** — Similar regions protest for similar structural reasons
3. **True contagion** — Protests in one place *cause* protests elsewhere

Approach: Discrete Spatial Hawkes Process

Hawkes processes decompose event intensity into:

- **Background rate** $\mu_{t,r}$: Exogenous events driven by structural factors
- **Triggered component**: Endogenous events caused by prior events

Spatial structure: We can compare different spatial kernels

- Local spread: Nearby regions matter more (finite θ)
- National spread: All regions weighted equally ($\theta = \infty$)

Literature: Hawkes (1971); adapted for discrete counts following Mohler et al. (2011), Schoenberg et al. (2019)

Research Design

Scope: Cross-Region Contagion

Two simplifying choices:

1. **Event definition:** Protests within 7 days at the same location $\rightarrow$ single event
   - Avoids double-counting continuation of the same protest
   - Focuses on distinct mobilization episodes

2. **Cross-region focus:** We model whether protests *elsewhere* predict protests *here*
   - Cleaner identification: controls for local persistence
   - Directly tests spatial diffusion across regions
   - *Limitation:* Cannot study within-region persistence (do protests here $\rightarrow$ more protests here?)

Model Specification

Baseline model (M0): Inhomogeneous Poisson with covariates only

$$E[Y_{t,r}] = \exp\left(\beta_0 + \beta_1 \log(\text{pop}_r) + \beta_2 \cdot \text{pov}_r + \beta_3 \cdot \text{CPI}_t + \gamma_{\text{year}}\right)$$

Spatial-temporal Hawkes model (M1): Adds cross-region contagion

$$E[Y_{t,r}] = \exp\left(\mathbf{X}'_{t,r}\boldsymbol{\beta} + \alpha \cdot \text{Cross}_{t,r}(\theta)\right)$$

where the cross-region exposure term captures **recent protests in other regions**:

$$\text{Cross}_{t,r}(\theta) = \sum_{s \neq r} w_{rs}(\theta) \cdot \underbrace{\sum_{\ell=1}^{30} Y_{t-\ell,s}}_{\text{past 30 days}}$$

Spatial Kernel Specification

Exponential decay kernel: $w_{rs}(\theta) \propto \exp(-d_{rs}/\theta)$

Intuition: Nearby regions matter more than distant ones

- θ = characteristic decay distance (km)
- At distance θ, influence drops to $\approx 37\%$ of its maximum
- At distance 2θ, influence drops to $\approx 14\%$
- At distance 3θ, influence drops to $\approx 5\%$

Key test: Is θ finite or infinite?

- $\theta < \infty$: Local spread — nearby regions matter more
- $\theta = \infty$: National spread — all regions weighted equally

Estimation Strategy

Estimation: Quasi-Poisson GLM with log link

- Accommodates overdispersion (estimated $\phi = 1.05$)
- Clustered inference not required: dispersion-adjusted SEs

Profile likelihood for θ:

1. Fix $\theta \in \{50, 100, 200, 300, 500, 750, 1000, 1500, 2000, 3000, 5000, \infty\}$ km
2. Estimate $(\boldsymbol{\beta}, \alpha)$ conditional on θ
3. Select $\hat{\theta}$ minimizing deviance

Hypothesis tests:

- $H_0^{(1)}$: $\alpha = 0$ (no contagion) — F-test comparing M0 vs M1
- $H_0^{(2)}$: $\theta = \infty$ (no distance decay) — LR test

Data

Data: ACLED Indonesia

Armed Conflict Location and Event Data Project

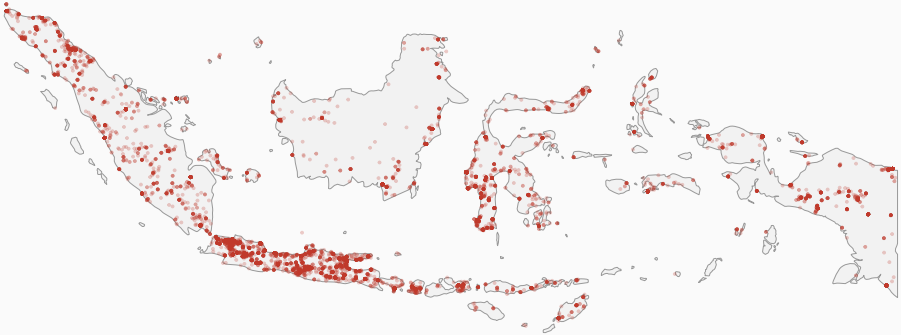
Sample:

- Period: January 2015 – October 2024
- Events: 12,576 protests
- Units: 434 districts (*kabupaten/kota*)
- Observations: 1.56 million district-days

Covariates:

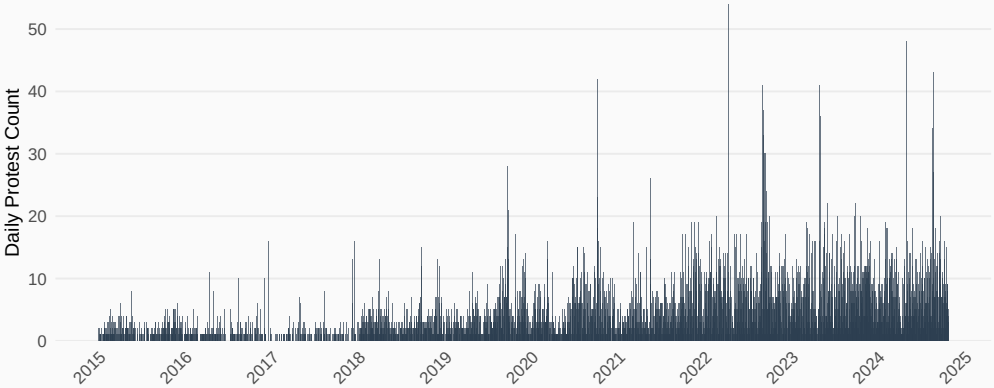
- $\log(\text{population})_r$ (district)
- Poverty rate_r (district)
- CPI_t (monthly, national)
- Year fixed effects

Spatial Distribution



Protests concentrate in Java (center); sparse in eastern Indonesia

Temporal Distribution



Clustering and temporal bursts consistent with self-exciting dynamics

Results

Test 1: Does Cross-District Contagion Exist?

	M0: Background Only	**M1: + Contagion** ($\theta = 100$ km)
Deviance	108,713	108,518
Parameters	13	14

F-test for nested models:

$$F = \frac{(D_0 - D_1)/1}{\phi} = \frac{(108713 - 108518)/1}{1.056} = 184.6$$

$$F(1, 1{,}557{,}187) = 184.6, \quad p < 10^{-41}$$

Result: Strong evidence for cross-district contagion

Coefficient Estimates

	Estimate	**SE**	z	p
Intercept	-19.03	0.902	-21.1	< 0.001
log(population)	0.636	0.012	54.5	< 0.001
Poverty rate	-0.021	0.002	-11.7	< 0.001
CPI	0.049	0.009	5.7	< 0.001
Cross-district (α)	**0.323**	**0.023**	**14.0**	**<0.001**
Year fixed effects included (2016–2024)				

Dispersion $\phi = 1.056$; $N = 1{,}557{,}200$ district-days

Interpretation: Multiplicative effect $\exp(\alpha) = 1.38$ per unit increase in cross-district exposure

Magnitude: Branching Ratio and Endogenous Fraction

Branching ratio (expected triggered events per original event):

$$\text{BR} = \sum_{s} \sum_{\ell} E[\lambda_s] \cdot (\exp(\alpha \cdot w_{rs}) - 1) = 0.11$$

- **Subcritical** (BR < 1): Process is stable, not explosive
- Each protest triggers ≈ 0.11 additional protests elsewhere

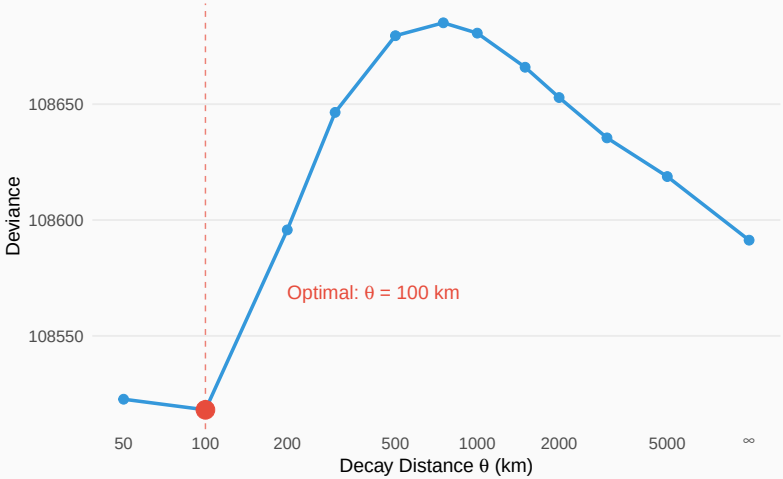
Endogenous fraction (share of events due to contagion):

$$\frac{\sum_{t,r} \left(\hat{\lambda}^{\text{full}}_{t,r} - \hat{\lambda}^{\text{background}}_{t,r} \right)}{\sum_{t,r} \hat{\lambda}^{\text{full}}_{t,r}} = 14.4\%$$

- $\sim$86% driven by background (exogenous) factors
- $\sim$14% attributable to cross-district contagion

Test 2: Does Distance Matter?

Profile likelihood over θ:



Test 2: Formal Hypothesis Test

Test: $H_0 : \theta = \infty$ vs $H_1 : \theta < \infty$

	$\theta = 100$ km	$\theta = \infty$
Deviance	108,518	108,591
$\hat{\alpha}$	0.323	1.022

Likelihood ratio test:

$$\text{LR} = \frac{D_\infty - D_{\hat{\theta}}}{\phi} = \frac{108591 - 108518}{1.056} = 69.1$$

$$\chi^2(1) = 69.1, \quad p < 0.001$$

Conclusion

Findings:

1. **Recent protests elsewhere predict protests today**
   - Protests in other districts over past 30 days $\rightarrow$ more protests here
   - $F(1, 1.56\text{M}) = 185$, $p < 10^{-41}$
2. **Effect is local, not national**
   - $\hat{\theta} = 100$ km; influence decays with distance
   - Reject uniform national spread ($p < 0.001$)
3. **Effect is modest but significant**
   - BR $= 0.11$ (subcritical); 14% triggered by cross-district contagion

Contribution: Discrete spatial-temporal Hawkes model; distinguishes local vs. national diffusion

Next Steps: Violent vs. Non-Violent Events

Question: Do violent and non-violent events have different mobilization potential?

Approach:

- Separate events by type (protests vs. riots/violence)
- Estimate cross-type contagion effects:
  - Do non-violent protests trigger more protests?
  - Do violent events trigger more violence—or suppress mobilization?
- Compare branching ratios across event types

Why it matters:

- Informs debate on tactical choice in social movements
- Tests whether violence escalates or dampens collective action

Thank You

Questions?

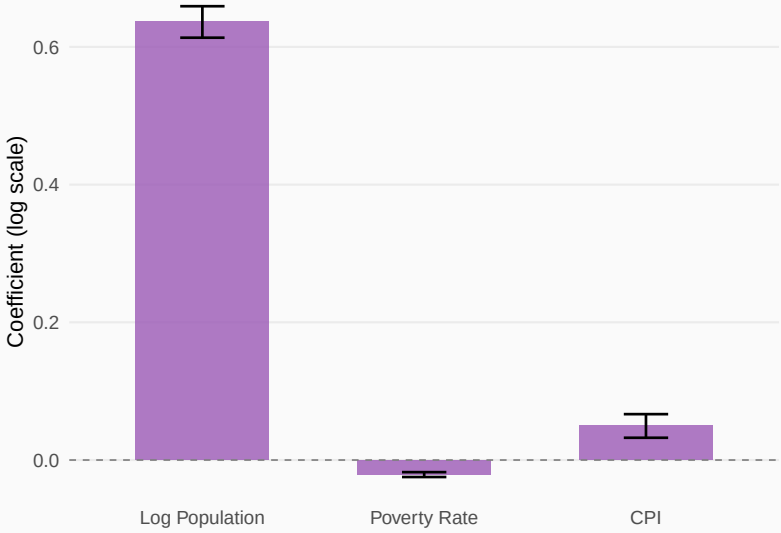
Appendix: Temporal Lag Structure

Model with weekly lag bins ($\theta = 100$ km):

	Estimate	**SE**	t	p
Week 1 (lags 1–7)	4.28	0.25	17.0	< 0.001
Week 2 (lags 8–14)	0.49	0.29	1.7	0.091
Week 3 (lags 15–21)	0.24	0.30	0.8	0.421
Week 4 (lags 22–30)	-0.42	0.24	-1.7	0.086

Effect concentrated in Week 1; fades rapidly thereafter

Appendix: Covariate Effects



Appendix: Profile Likelihood Detail

θ (km)	Deviance	$\hat{\alpha}$	SE($\hat{\alpha}$)	t
50	108,523	0.229	0.016	13.9
100	**108,518**	**0.323**	**0.023**	**14.0**
200	108,596	0.385	0.036	10.7
300	108,647	0.368	0.047	7.9
500	108,680	0.336	0.060	5.6
1000	108,681	0.420	0.076	5.5
2000	108,653	0.645	0.086	7.5
∞	108,591	1.022	0.095	10.7

Note: Minimum deviance at $\theta = 100$ km